Enroll. No._____

**MARWADI UNIVERSITY**

**Faculty of Technology**

**Information and Communication Technology**

**B.Tech** **SEM: VI** **Mid 2 Exam** **April 2024**

**Subject: - Artificial Intelligence (01CT0616)** **Total Marks:-30**

## 1. What is Artificial Intelligence?

A) Database. B) Simulation of human intelligence

C) Hardware. D) Network **Answer:B**

## 2. What does NLP stand for?

A) Network Layer Protocol B) Natural Language Processing

C) Neural Language Program D) Numeric Logic Processing **Answer:B**

## 3. What is Machine Learning?

A) Manual coding B) Learning from data

C) Hardware design D) Networking **Answer:B**

## 4. What is a dataset?

A) Single value B) Collection of data

C) Software D) Tool **Answer: B**

## 5. What is tokenization?

A) Encryption B) Splitting text

C) Storage D) Compression **Answer: B**

## 6. Which of the following best describes supervised learning in Artificial Intelligence?

A) Learning with labeled data B) Learning without labels

C) Learning using rewards D) Learning randomly **Answer:A**

## 7. Which technique is commonly used to measure similarity between two text documents in NLP applications?

A) Sorting B) Cosine Similarity

C) Searching D) Hashing **Answer: B**

**8. What is the purpose of TF-IDF in text processing and information retrieval systems?**

A) Image processing B) Keyword extraction

C) Data storage D) Encryption **Answer:B**

**9. Which of the following correctly defines reinforcement learning in Artificial Intelligence systems?**

A) Learning with labeled data B) Learning through rewards and penalties

C) Learning without any data D) Learning using images only **Answer:B**

**10. What is the main role of an agent in an Artificial Intelligence system interacting with an environment?**

A) Store data B) Make decisions and perform actions

C) Compile programs D) Display results **Answer: B**

**11. Which library is widely used for Natural Language Processing tasks in Python programming?**

A) NumPy B) NLTK

C) Pandas D) OpenCV **Answer: B**

**12. What is the primary function of stopword removal during text preprocessing in NLP tasks?**

A) Remove important keywords B) Remove common words like “is”, “the”

C) Add extra words D) Encrypt text **Answer:B**

**13. Which concept in game theory refers to a situation where no player benefits by changing their strategy alone?**

A) Zero-sum game B) Nash Equilibrium

C) Heuristic D) Tokenization **Answer:B**

**14. What is the function of the open list in the A search algorithm used in pathfinding problems?**

A) Store explored nodes B) Store nodes to be explored

C) Store final path D) Store heuristic values **Answer:B**

**15. What is the purpose of Optical Character Recognition (OCR) in AI-based systems and applications?**

A) Audio processing B) Text extraction from images

C) Video editing D) Networking **Answer:B**

**16. In reinforcement learning, what does a reward signal represent when an agent interacts with an environment and takes actions?**

A) Punishment for incorrect actions B) Feedback indicating success or failure

C) Cost of computation D) Random output **Answer: B**

**17. What is the key difference between supervised learning and reinforcement learning in terms of how feedback is provided to the model?**

A) Supervised learning uses rewards

B) Reinforcement learning uses labeled data

C) Reinforcement learning uses interaction-based feedback

D) No difference

**Answer:C**

**18. In Natural Language Processing, what is the role of tokenization when preparing raw text data for further processing?**

A) Encrypt text B) Split text into smaller units like words

C) Store data D) Compress files **Answer:B**

**19. What is the purpose of cosine similarity when comparing two documents in AI-based answer evaluation systems?**

A) Measure distance B) Measure similarity between text vectors

C) Sort documents D) Encrypt data **Answer: B**

**20. Which technique is used to extract important keywords from a dataset by considering both frequency and uniqueness of terms?**

A) Sorting B) TF-IDF

C) Searching D) Hashing **Answer: B**

**21. What is the role of the heuristic function in the A search algorithm when finding the shortest path in a graph?**

A) Exact cost calculation B) Estimated cost to goal

C) Random value D) Final output **Answer: B**

**22. In the A search algorithm, what is the significance of combining path cost and heuristic value for node evaluation?**

A) Random search B) Optimal path finding

C) Data storage D) Encryption **Answer: B**

**23. In game theory, what does a zero-sum game indicate about the relationship between players' gains and losses?**

A) All players win B) One player's gain equals another's loss

C) No loss D) Random outcome **Answer: B**

**24. What is Nash Equilibrium in a competitive game scenario where multiple players choose strategies independently?**

A) Best mutual strategy B) Random decision

C) Worst case D) No solution **Answer: A**

**25. In an AI-based answer evaluation system, how is similarity between student answer and model answer typically calculated?**

A) Sorting B) Cosine similarity

C) Searching D) Encoding **Answer: B**

**26. What is overfitting in machine learning models when trained on a limited dataset with high accuracy?**

A) Poor performance B) Model fits training data too well

C) No learning D) Random output **Answer: B**

**27. What is underfitting in machine learning when a model fails to capture patterns from the training data?**

A) Perfect model B) Poor learning

C) High accuracy D) Complex model **Answer: B**

**28. In NLP preprocessing, why is stopword removal important when analyzing large text datasets?**

A) Increase noise B) Reduce unnecessary common words

C) Add complexity D) Encrypt data **Answer: B**